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" On my honor, I have neither given nor received any unauthorized aid or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues. "

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(3) b-) $Hu = (I - 2uu^T)u = u - 2uu^T u = u - 2u$

$= -u$ ✓

$= \boxed{-1}$

a-) $H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uuu^T uu^T$ because,

$u^T u = 1$, $H^2 = I$

$H = H^T$ so, $HTH = I$

Hence, H is an unitary/orthogonal matrix.

c-) $Hv = (I - 2uu^T)v = v - 2uu^T v = v$ Therefore,

$\lambda = 1$

u tarafından yayılan bir uzayın ortogonal tümleyeni $n-1$ boyutuna sahip olmaktadır. Bu sebeple, bu özdeğer sahip $n-1$ ortonormal öz vektörler bir temele sahiptir. u özvektörünü ekleyerek, H'nin köşegenleştirilebilir (diagonalizable) olduğunu bulabilirsiniz.

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(2) (a) x=5y line in 3D and can intersect a shape with 3D volume. That is why, it is three-dimesional. We can talk about volume in more dimesions. Nonetheless, it happens in at least three-dimensions.

The points is in 1D. The junction of two points becomes a line and is two-dimentional. If we combine the two lines, we can get volume in 3 dimentions.

(b) It may become a rectangular prism, if two planes converge.

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(4) (b) $\frac{d}{dt}\begin{bmatrix} v \\ u \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$

$0 \quad 0$ alt alta yazdığımız zaman sıfır olmak zorunda, çünkü su kaybı ya da sıvı kaybı yoktur.

$|A - \lambda I| = \begin{vmatrix} -1-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix}$ $\quad$ Trace $= \lambda_1 + \lambda_2 = -2$

$\lambda_1 = 0$ (dependent eqs) $\quad \lambda_2 = -2$

$(-1-\lambda)\cdot(-1-\lambda) = 1 + 2\lambda + \lambda^2$

karakteristik denklem ✓

$\lambda_1 = 0 = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \quad x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad \lambda_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

↳ (eigen vektor)

$\lambda_2 = -2 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

↳ eigen vektor

$v(t) = s_1 e^{\lambda_1 t} c_1 + s_2 e^{\lambda_2 t} \cdot c_2$

$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{\lambda_1} \cdot c_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t} \cdot c_2$

initially,

tank A $v(0) = 10c$

tank B $w(0) = 40c$

$\begin{bmatrix} 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} c_1 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} c_2$

for $t(0)$

$40 = c_1 - c_2$

$10 = c_1 - c_2$

$50 = c_1$ $\quad c_1 = 25$

$c_2 = -15$

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4. Sorunun devamı

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 25 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t}(-15)$$

for $A = 25 + (-15)$: $e^{-2} = \%.45$

for $B = 25 + 15e^{-2} = \%.55$

(c)

$$v(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} 25 + \begin{bmatrix} 1 \\ -1 \end{bmatrix} e^{-2t}(-15)$$

$t = \infty$

if it goes infinity, the right hand side is 0

Therefore;

$A = \%50$

$B = \%50$ > remains liquit

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